Tutorial-05

Q) A Healthcare system ~~wants~~ wants to pred whether a patient has diabetes using clinical attributes.

Input
- features
- Age
- BMI
- Blood Glucose level
- Blood pressure.

target variable: Diabetes (1=Yes, 0=No)

Age	BMI	Glucose	BP	Diabetes
45	28	160	85	1
30	22	110	70	0
50	31	180	90	1
25	20	95	68	0
40	26	140	80	1

Questions

1) Identify the classification type of this problem.
2) write the logistic Regression hypothesis function
3) Given predicted probabilities, compute the log loss value for 3 iterations.

1A Target value is diabetes ({0,1}), so there is a binary classification because output has only two labels that is Yes or (no) 0 or 1.

2A $z = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4$

$= w_0 + w_1 \text{Age} + w_2 \cdot \text{BMI} + w_3 \cdot \text{Glucose} + w_4 \text{BP}$

$$\text{Sigmoid} = \frac{1}{1+e^{-z}}$$

$J = -[y \cdot \log(\hat{y}) + (1-y)\log(1-\hat{y})]$

Normalization is a process of scaling numerical features to common range, usually [0,1] or [-1,1].

$$x = \frac{x - x_{minvalue}}{x_{max} - x_{min}}$$

After Normalization

	age	BMI	Glucose	BP	y
1)	0.8	0.73	0.76	0.77	1
2)	0.2	0.18	0.18	0.09	0
3)	1	1	1	1	1

first sample

$x(0.8, 0.73, 0.76, 0.77)$ $y=1$

$z = 0.01 + 0.01(0.8) + 0.01(0.73) + 0.01(0.76) + 0.01(0.77)$

$z = 0.0406 = \sigma(z) = \frac{1}{1+e^{-0.0406}}$ 0.5101

$J = -\log(1-\hat{y})^{1-y}$ $y=0$

$= -\log(1-0.5101)$

$= 0.3$

$error = \hat{y} - y = 0.5101 - 1$

$= -0.4899$

$w_{new} = w_{old} - \eta(Error) \cdot x$

$= 0.01 - 0.1(0.4899)0.8$

$= 0.049192$

$w_2 = 0.01 - 0.1(-0.4899)(0.73)$

$w_2 = 0.04576$

$w_3 = 0.01 - 0.1(-0.4899)(0.76)$

$= 0.04723$

$w_4 = 0.01 - 0.1(-0.4899)(0.77)$

$= 0.04772$